Analytical Cume – January 25, 2013

Based on the original Paper by Velmer Fassel
"Current and Potential Applications of Inductively Coupled Plasma (ICP)-Atomic Emission Spectroscopy (AES) in the Exploration , Mining, and Processing of Materials. Pure & Appl. Chem., Vol. 49, pp. 1533-1545, 1977.

1. Atomic Emission (20 points):

a. Give a freshmen level explanation of atomic emission.

b. It is also possible for a cation to emit a line spectra, producing another set of lines, separate from the atoms. Would the cation spectra be at shorter or longer wavelengths from that of the atom? To get credit you must support your answer with some freshman chemistry concepts.

2. Compare and contrast atomic absorption spectroscopy(AAS) with atomic emission (AES). (30 points)

a. Velmer Fassel, claims that AES fell from favor over AAS because of what reasons?

b. Velmer Fassel, claims that AES is now surpassing AAS in popularity because of what reasons?

c. Which technique can do most analyses per hour? And why is it so much faster?

3. The plasma: (25 points)

a. Define plasma.

b. Draw a diagram of the, induction coil, the resulting magnetic field, and the eddy currents within the plasma source.

c. Near the plasma the temperatures exceed 9,000 K. What mechanism in the plasma causes temperatures to be so high?

d. Why does the plasma take-on a donut shape? i.e. why does it not fill-up the whole space between the coils?

e. List three reasons why the plasma source is superior to standard flames.

4. Inter-element Interferences (10 points)

The chemistry going on in a flame has a lot to do with the spectral output of each atoms/ions emission. Consider the following:

$Ca^{2+} \leftrightarrow Ca^{3+} + 1e^-$	Ionization energy (kj/mol)	4,912
$Na^{+} \leftrightarrow Na^{2+} + 1e^-$	Ionization energy (kj/mol)I:	4,560

If you are analyzing for calcium (II) in the presence of excess sodium(I), how would sodium ion concentration affect the intensity of the lines from calcium(II)?

5. Instrumental Aspects (15 points)

In Velmer's day, What component in the ICP-AES instrumentation is most responsible for the "stray light" component of the background?